RAGHEB FAISAL

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Portfolio Link-https://drive.google.com/drive/folders/1V2W1GYtyobds1gDaWc1-15XAUKrJMuHW?usp=sharing

EXPERIENCE

Inductoheat Inc, Design Engineer, Madison Heights, MI, USA
September 2023 – Present

- Design and develop custom Induction Coils to heat treat a variety of automotive parts to customer specifications.Some examples include Differential Ring gears,Truck Channels,Differential Case Carriers,Drive Shafts etc.
- Coordinating with the Manufacturing department in order to optimize existing and new design as well as ease of manufacturing.This ranges from material selection to design changes based on manufacturing/development input.
- Use **SolidWorks** and **Solidworks PDM** to create/store CAD Models of induction coils, buswork, and quench assemblies.
- Responsible for creation of Engineering Drawings packages and ensuring their compliance with company drafting standards using **Solidworks** and **AutoCAD**.
- Responsible for creating Bill of Materials,Engineering Release Orders(as needed) and Engineering Change Orders(as needed) for new designs and existing Blueprint Builds using **Symix** MRP.
- Some examples of projects include-
  - Single Shot Induction Coils- Mainly used for heat treatments of differential shafts,Stator Housing and Differential Case carriers.
  - 60 Hz Core Type Coils- Used for heat treatment of multiple ring gear configurations.
  - I.D. Coils- Used for heat treating small Differential Case windows and small regions in general.
  - Profile Coil-Used to heat treat various truck channel rails and varying thickness.Also designed an associated winding mandrel for making the same coil.
  - Sharp-C harden coils(CAD Modelling and Drawing creation only)-Used for heat treatment of Crank and Camshafts.
- Part of the Internal audit team at Inductoheat. Responsibilities include full-scale audits of the various departments at Inductoheat to ensure that we meet ISO 9001:2015 and Inductoheat Internal standards.

TSS Inc, Mechanical Design Engineer, Warren, MI, USA
June 2022 – September 2023

- Use **SolidWorks** and **Solidworks PDM** to create and store detailed 3-D models ,Flat Patterns(Sheet-Metal Design) electrical wiring diagrams, and engineering drawings of existing and new product designs while maintaining Geometric Dimensioning and Tolerancing.
- Produce Engineering Drawings, Bill of materials (BOM), and product pricing(quote) for the same using **SolidWorks**, and **Microsoft Excel** respectively.
- Collaborated with the engineering team to develop a modified version of the Rain Bar, utilizing Solidworks to create engineering drawings and a working prototype; successfully reduced foam accumulation within the bar by 40%.
- Led efforts with the production team to optimize the CTA Tire Foamer, reducing required components by 28%; fine-tuned spray nozzle dimensions to achieve a uniform spray pattern for mid-size vehicles.
- Developed and implemented multiple design and manufacturing projects such as the Lightweight Canopy, Valet Fixture Sign, Holographic Projector, Triangle I-Conformational Sign, Rotating Bucket Sign, and Titan arches(Manufacturing only).
- Collaborate with the Sales department to address customer queries and complaints regarding product installation.

National Automotive Design League, CAD Engineer, India
Oct 2020 – Feb 2021

- Led development efforts as a CAD Engineer in Team Karnataka and collaborated with a team of professionals to create a comprehensive 3-D design of an Electric Motorcycle using **PTC Creo** and **SolidWorks CFD**.
- Derived the final value of drag coefficient, the X and Y forces acting on the fairing, the required brake torque and brake force for both the front and rear wheels along with the associated stopping distance.
- Achieved a safety factor of 1.5 and 2 for the brake and torque forces respectively and validated the same using **Ansys Mechanical**.

- Led an in-depth Fairing flow analysis case study; successfully managed to reduce the drag coefficient of the fairing (with rider) to 0.255 using **Solidworks CFD**.
- Created detailed 3-D models along with technical documentation and the final PowerPoint presentation of the disc brake assembly, tire and hub assembly, Fairing, and Transmission assembly with **PTC Creo**.

ACADEMIC PROJECTS

Presoak and Rinse Pneunet Actuator, Lawrence Technological University, Southfield, MI
April 2023 – June 2023

- Created a 3D CAD model for a Elastosil pneunet actuator using **Solidworks**. The prototype main function is to function as a Chemical Applicator to be used with the Carwash industry.
- The actuator would be mounted on an application arch and connected to a water and chemical pump.On supplying inlet pressures of upto a maximum of 7 psi the actuator would enlarge and deform around the shape of the vehicle.Once the supply is turned off the actuator would shrink to its original shape.
- In order to validate this prototype, Simulations were run on a half model (License restrictions) in order to find out the deformation (in mm), Elastic Strain and Equivalent Stress(Pa) using **Ansys Fluent**.

CFD Modeling for Inlet Manifolds, Engine Radiators, and Butterfly Valves, Lawrence Technological University, Southfield, MI
May 2021 – May 2023

- Compared four different configurations of Engine radiators based on Pressure drop, Temperature drop, and Tube surface uniformity using **Ansys Fluent**. The results were supported with detailed contours, plots, and surface reports.
- Conducted a case study on a shell-and-tube heat exchanger in order to confirm the advantages of counterflow as opposed to parallel flow in terms of heat dissipation. Results were supported with detailed plots, contours, and reports all generated using **Ansys Fluent**.
- Performed a detailed analysis of an impeller and housing assembly where we compared the pressure rise on the inlet side at 100%,75%, and 50% of the maximum mass flow rate. Created a detailed report comparing the efficiency, pressure rise, and flow pattern on all 3 configurations using **Microsoft Excel** and **Ansys Fluent**.

Suspension System Modeling for Sports and Luxury Cars, Lawrence Technological University, Southfield, MI
August 2021 – December 2021

- We created a **Simulink** model for a half-car suspension using **MATLAB**. The model was tuned for a luxury and sports setup respectively.
- Selected the suspension values and validated the same using the **Simulink** model in order to provide maximum comfort and minimum travel when driven on a sinusoidal and random road profile.
- Using **Simulink**, Managed to reduce the velocity and acceleration variance (for the first 10 seconds) to 1*10^(-4) m and 2*10^(-5) m for the luxury and sports models respectively.
- Created the final technical report in which the **Simulink** plot results for the Acceleration, Displacement, and Velocity were included along with the Equations of Motion (Derived) and requisite calculations.

TECHNICAL SKILLS

- Catia V5
- MATLAB (Simulink)
- ANSYS Fluent
- SolidWorks
- AutoCAD
- Microsoft Excel
- Fusion 360
- PTC Creo

EDUCATION

Lawrence Technological University, Southfield, MI, USA
August 2021 – December 2023

Master of Science in Automotive Engineering, **GPA: 3.87**